Namal University, Mianwali

Department of Electrical Engineering
EE 361L: Control Systems Lab

Lab 13
Controlling a Lego Robot using PID controller

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Marks Obtained			
Instructor's signature			

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Objective:

The purpose of this lab is to use all the skills we practiced in previous labs and control a Lego Mindstorms EV3 differential derive robot. We have implemented open-loop and close-loop (PID) controller to derive the robot in straight line. Our group has to work on closed loop (PID).

Apparatus:

- Computer with installed MATLAB software
- Components of Robots
- EV3 brick

Introduction:

In a vehicle using independent wheel control, applying the same power to each wheel generally does not result in the vehicle moving straight. This is caused by mechanical and surface differences experienced by each of the wheels. To reduce deviation in the vehicle heading, a better approach is to use a closed-loop controller which adjusts the power applied to two motors based on difference in their rotations. One such controller is a well-known proportional-integral-derivative (PID) controller. Proportional-integral-derivative control is a basic control loop feedback mechanism. The controller minimizes the difference between the measured and the desired value of a chosen system variable by adjusting the system control inputs. This example shows you how to simulate the controller using a simple plant model, first with no feedback control (Open-Loop Control), and then with feedback control (Closed-Loop Control). So there were two groups formed one was working with open loop while one was working with closed loop, so out team was working on closed loop.

Task 3 Simulating Close-Loop system

1. Open the **model** by running following command in MATLAB command window and observe two subsystems in the model.

```
ev3_drive_closedloop
```

2. Double-click **PID Controller** subsystem. Notice that the P control is used to synchronize two motors when the vehicle runs straight. Also, notice that during a turn, no synchronization is applied as shown in Figurews below.

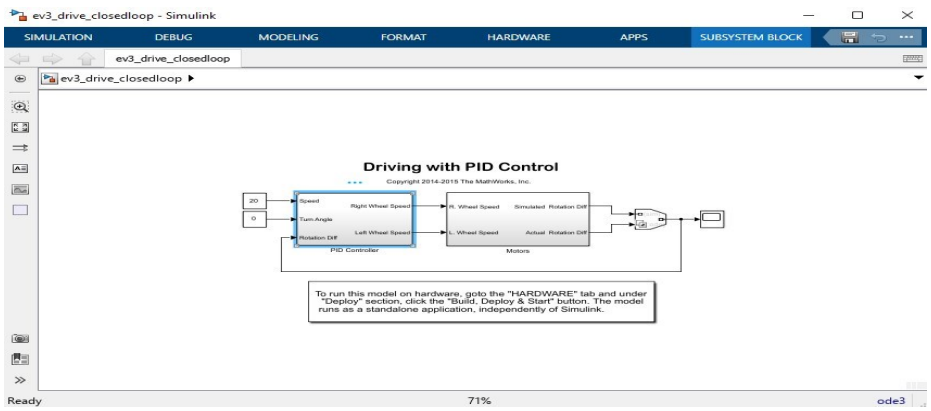


Figure 12: Close loop Control

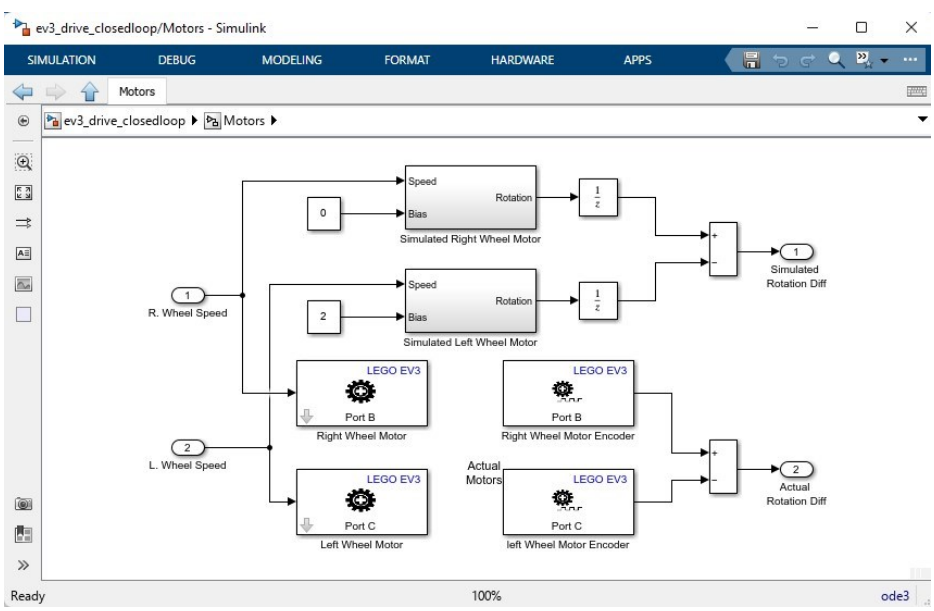


Figure 13: Internal Structure of Motor Block

3. Click **Run** button in the Simulink toolbar. Click the **Scope** block and observe that the **Encoder Output Mismatch** remains close to zero. This indicates that the vehicle will skew less compared to the **Open Loop Control** model.

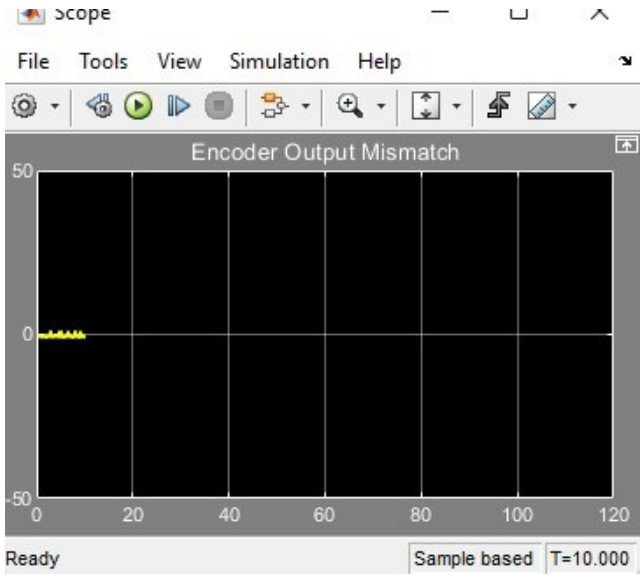
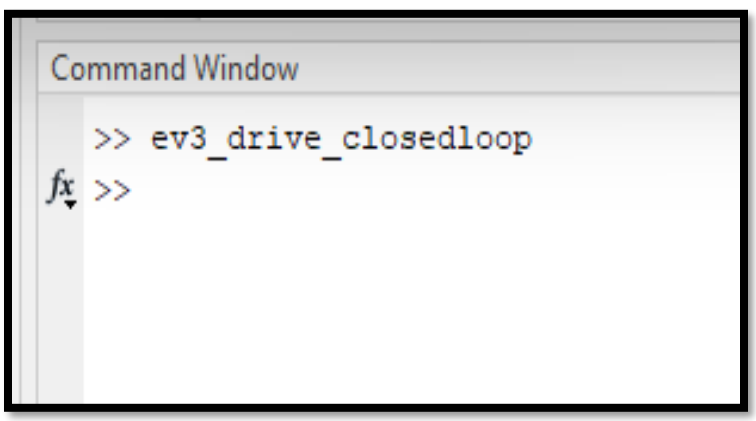


Figure 14: Encoder Output

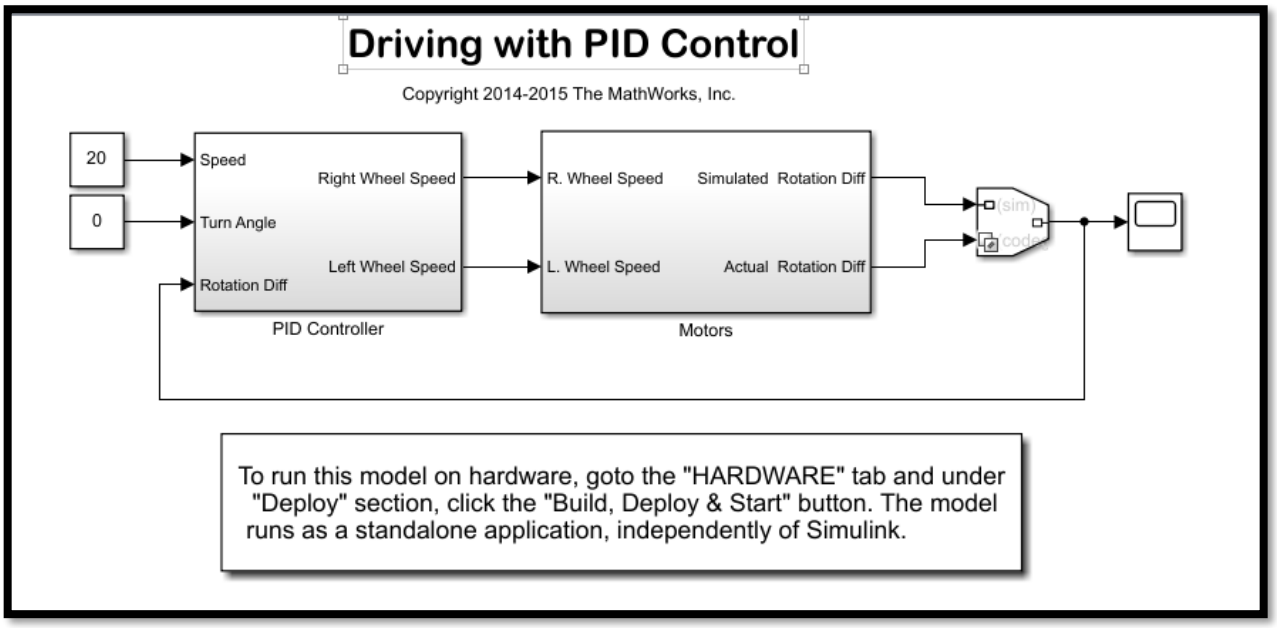
4. In the **Hardware** tab of your Simulink model, click **Build, Deploy & Start**. The model runs on LEGO MINDSTORMS NXT hardware and the vehicle starts moving.

5. Notice that the path of the vehicle is straight, as predicted by the simulation result above.

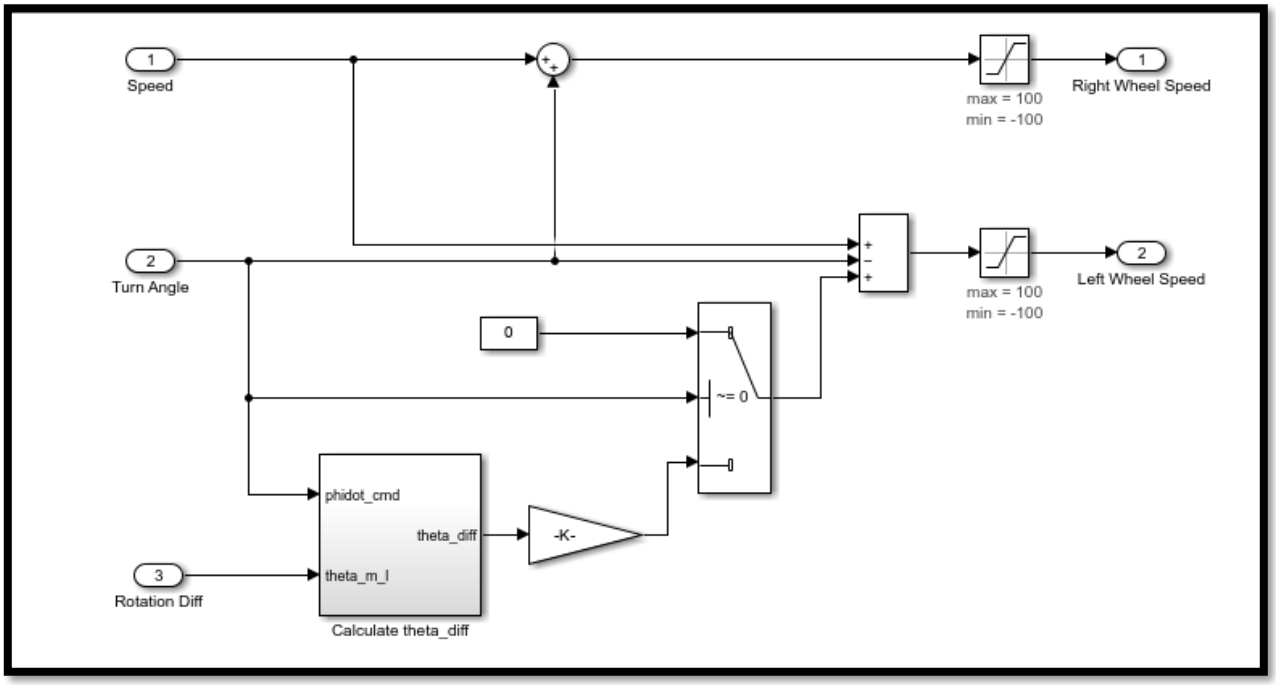
Closed Loop



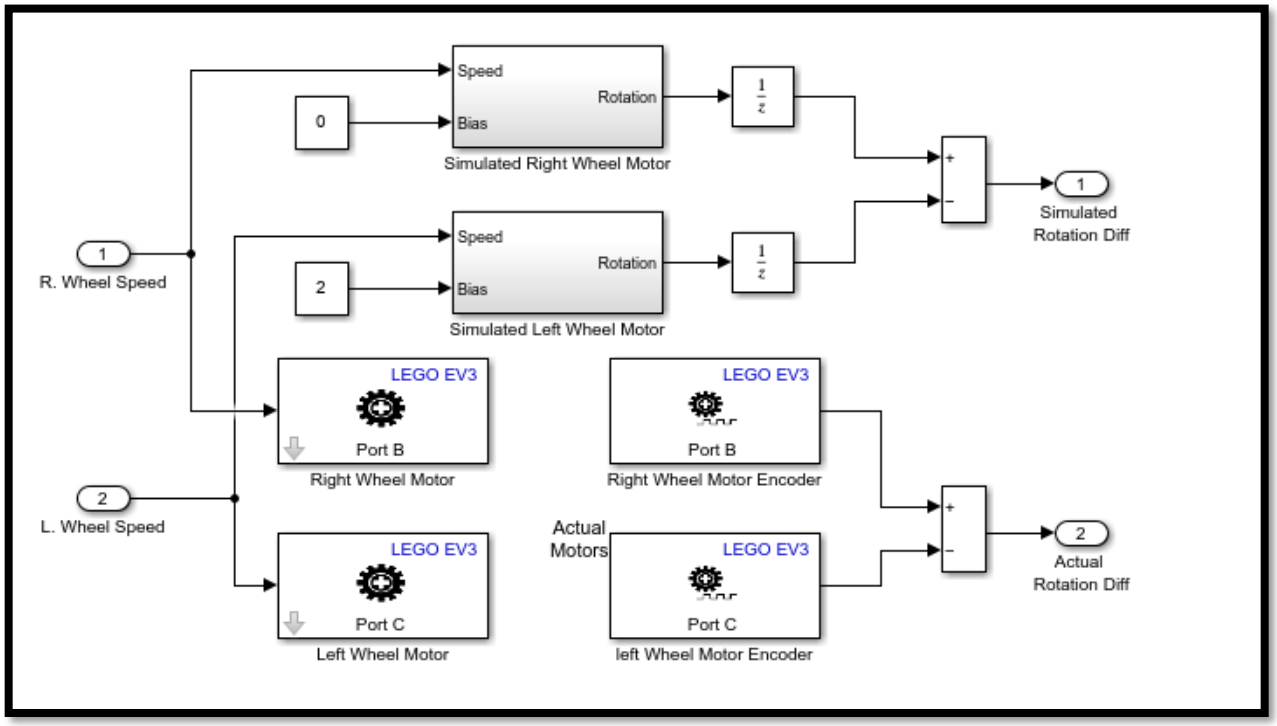
Closed Loop Simulation:



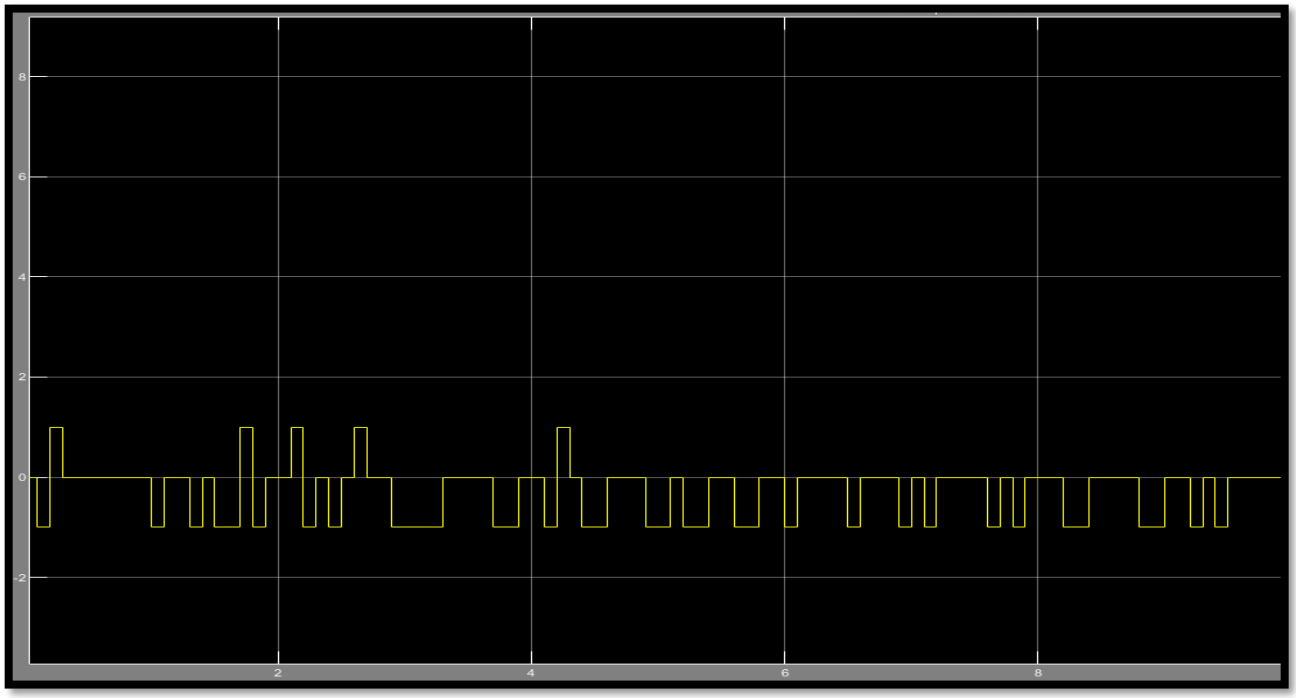
Closed Loop Simulation



Simulink of Closed Loop:



Simulation Graph of Closed Loop



Conclusion:

In this lab, basically we have designed a Lego Robot using PID Controller. We have done the implemented the programming of closed loop which is quite simple. We just have to give a command "ev3_drive_closedloop" in MATLAB Window. Then we burnt that code on EV3 in Lego robot. In this the element of feedback was involved. Now the robot will take feedback during its movement and determine any error (if caused) and will resolve that issue because of the feedback. So in case of closed loop the robot will move in straight line. Overall, this was a very good lab full of practical knowledge of building a Lego Robot from scratch. We have learnt the assembling of different components to build a robot. Then also we learnt that how simulation on MATLAB , or Simulink and hardware are associated with each other and how we can practically implement our simulation work on hardware.

Lab Rubric							
Domain	**Performance indicator**	**Excellent (10)**	**Good (7-9)**	**Satisfactory (4-6)**	**Unsatisfactory (13)**	**Poor (0)**	**Allocated Marks**
Cognitive (20%)	Lab Demonstration (C)	All Outcomes are critically analyzed in a meaningful way.	Outcomes are partially analyzed / need minor improvement.	Outcomes are partially analyzed but need more emphasis regarding conclusion.	Outcomes are not properly analyzed/student understanding is weak.	Outcomes are not analyzed	
Psychomotor (60%)	Task Completion (P)	All tasks completed correctly in given time	Most tasks completed correctly in given time	Some tasks completed correctly in given time	Most tasks incomplete or incorrect	All task incomplete or incorrect	
	Code (P)	Correct Code	Correct code but without explanation	Slightly incorrect code	Almost Incorrect code	Code not Submitted	.
	Output (P)	Output correctly shown with proper labeled	Most Output/Figures/Plots displayed with proper labeled	Some Output/Figures/Plots displayed with proper labeled	Most of the required Output/Figures/Plots not displayed	Output/Figures/Plots not displayed	
Affective (20%)	Discipline (A)	Perform as an individual or in a group with enthusiasm	Actively perform as an individual or in a group	Partially perform as an individual or in a group	Does not perform any task individual or in a group	Have not attend the lab session	